Design of Enterprise IoT

Tutorial 4

IoT Computing Connectivity

Agenda

- Reminder: Week 5 quiz (available till Friday 27 March 23.59)
- Recap of Week 4's lecture
  - Fog & Cloud Computing
- Tutorial activity
  - Read the use case
  - Continue working on your IoT connectivity proposal
  - **This week**: Determine the connectivity between IoT devices and network devices within the edge/fog layer and to the cloud.

➢ *You are encouraged to work in groups*

Fog vs. Cloud Computing

Requirement	Cloud computing	Fog computing
Latency	High	Low
Delay jitter	High	Very low
Location of service	Within Internet	Network edge
Distance between data sources and consumers	Multiple hops	Single hop
Location awareness	No	Yes
Geo-distribution	Centralised (data centre)	Distributed
# of server nodes	Medium – Large (homogeneous)	Large – Very Large (heterogeneous)
Support for mobility	Limited	Yes
Data analytics	Data at rest	Data in motion
Connectivity	Wired line	Wireless

(Rayes and Salam, 2022)

Tutorial Activity

IoT Use Case: Intelligent Port System for Fraud Detection



Photo credit to Smart Port Solution

Read the use case and determine IoT connectivity in the fog and cloud layers for the proposed solution

Intelligent Port System for Fraud Detection

Port operations play a crucial role in global trade, handling vast amounts of cargo daily. However, fraudulent activities, such as cargo theft, pose significant financial and operational risks. According to China Police Daily, between 2013 and 2017, Chinese Public Security Traffic Management Authorities cracked over 4,000 criminal cases related to port logistics (Song et al., 2020) . Cargo theft is often considered a "silent crime" because losses frequently go unreported and may only be discovered weeks or even months later.

One common cargo theft tactic involves manipulating a truck's empty weight before entering the port. Fraudsters add materials like liquid tanks, cement, or scrap to artificially increase the truck's weight during the initial weigh-in. After getting the empty truck weighed, these materials are secretly removed, creating a weight discrepancy that allows the truck to load excess cargo undetected. Such fraudulent practices lead to financial losses, regulatory violations, and operational inefficiencies.

Traditional fraud detection methods rely heavily on manual inspections, post-event investigations, and rule-based monitoring systems, which struggle to detect anomalies in real time. These limitations highlight the need for an intelligent port system that enhances security, improves efficiency, and minimizes fraudulent activities. IoT technology offers a promising solution by enabling real-time tracking, automated data collection, and intelligent analytics, allowing ports to proactively detect fraud and optimise operations.

Song, R., Cui, W., Vanthienen, J., Huang, L., & Wang, Y. (2022). Business process redesign towards IoT-enabled context-awareness: the case of a Chinese bulk port. *Business Process Management Journal*, *28*(3), 656-683.

Proposed IoT Solution

An IoT-based intelligent port system leverages advanced IoT technology and data analytics to enhance security and efficiency. Key components include:

- **Smart Weighbridge Systems:** Infrared sensors and weight sensors detect improper truck positioning and compare real-time weight with historical records, ensuring accuracy in cargo measurement.
- **Environmental Context Integration:** Weather sensors provide real-time rainfall data to detect fraudulent weight manipulation involving hidden water tanks.
- **GPS-Based Route Monitoring:** GPS tracking ensures trucks follow designated routes, detecting unauthorised stops or route deviations that may indicate fraud.
- **IoT-Enabled Cargo Surveillance:** Smart cameras, RFID scanners, and environmental sensors verify cargo handling, preventing unauthorised cargo movement.

Activity

Continue with the communication/network technologies you selected last week for the above Intelligent Port System for the Fraud Detection scenario. Determine the connectivity between IoT devices and network devices within the fog layer and to the cloud.

Consider the following questions:

- Determine the connectivity between IoT devices and network devices in the **edge/fog layer**
  - Is an edge/fog layer required? Why?
  - Which network device(s) should handle computing across the edge/fog layer?
  - What data is transmitted between the IoT devices and network device(s)?
  - What data processing and/or analysis tasks should be performed in the edge/fog layer?
  - If the fog needs to connect with a user application, what is the purpose of the connection?
  - If the fog needs to connect with an existing system, what is the purpose of the connection?

Activity (cont'd)

Continue with the communication/network technologies you selected last week for the above Intelligent Port System for the Fraud Detection scenario. Determine the connectivity between IoT devices and network devices within the fog layer and to the cloud.

Consider the following questions:

- Determine the connectivity to the **cloud** layer
  - Is a cloud layer required? Why?
  - What data should be transmitted to the cloud? Also, specify if the data is transmitted from IoT devices or edge/fog layer devices.
  - What data processing and/or analysis tasks should be performed in the cloud?
  - If the cloud needs to connect with a user application, what is the purpose of the connection?
  - If the cloud needs to connect with an existing system, what is the purpose of the connection?

Port Access Control

- Verify vehicle access to port and determine if or not to grant the access
- Trigger gate operation immediately
- Generate real-time alerts for unauthorised vehicles
- Store vehicle access history for vehicle arrival pattern analysis
- Analyse access-related anomalous behaviour



Smart Weighbridge

- Validate truck position
- Verify and record valid weight readings
- Trigger alert upon detecting invalid weight measurements
- Log truck weighing history
- Discovery truck weighing patterns
- Identify frequency and causes of invalid weighing





GPS tracking





